SN APPLIED SCIENCES · CASE STUDY

State E-Waste Management Tracker

A Scorecard-Based Case Study

Uttar Pradesh · FY 2026–27 · Aligned to the E-Waste (Management) Rules, 2022

72/100	6	7
State Score — GOOD band	Districts Tracked	Data Categories Audited

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Abstract

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This project builds a 100-point scorecard tracking e-waste generation, collection, formal recycling, EPR compliance, informal-sector integration, environmental and health safeguards, and awareness outreach across six districts of Uttar Pradesh — Lucknow, Kanpur Nagar, Ghaziabad, Gautam Buddh Nagar, Varanasi, and Agra. Every underlying formula was audited and validated prior to scoring, ensuring the results reflect genuine district performance rather than calculation error. The resulting state score is 72 out of 100, placing Uttar Pradesh in the GOOD performance band under the scorecard's classification system.

72 / 100
Overall State Score

GOOD (65–84 band)

6 of 7
Formula errors found & corrected

1 Introduction

- E-waste is the fastest-growing waste stream in India, driven by rising electronics consumption and shortening device lifecycles.
- India's E-Waste (Management) Rules, 2022 mandate Extended Producer Responsibility (EPR), formal collection targets, and authorized recycling channels.
- Despite these mandates, district-level performance data remains fragmented across departments, with no unified, live scoring mechanism to track progress.
- This tracker consolidates that data into a single, auditable scorecard — closing the visibility gap between policy targets and on-ground outcomes.

[INSERT IMAGE: E-waste collection / recycling facility photo]

1.1 Objectives

01

Consolidate

Bring together district-level generation and collection data into one structured workbook.

02

Validate

Audit and correct every scoring formula before results are treated as final.

03

Score

Compute a weighted, 100-point state score across seven performance categories.

04

Visualize

Present findings through an interactive dashboard for ongoing monitoring.

2 Methodology

Workbook Structure — 7 Linked Sheets

- Generation & Inventory
- Collection Log
- Recycling Infrastructure
- Compliance Tracker
- Informal Sector Hotspots
- Environmental & Health Monitoring
- Awareness Programs

Scoring Formula

Weights: 15 / 20 / 20 / 15 / 10 / 10 / 10 = 100

Achievement % = Achieved ÷ Benchmark, capped at 100%

Data Validation Step

- 6 formula errors identified across the workbook prior to finalizing the score
- Incorrect column references corrected in cross-sheet lookups
- Average-vs-sum mismatches resolved in aggregation formulas
- Overly strict text-matching logic relaxed for compliance status checks
- Every category re-verified against source data after correction

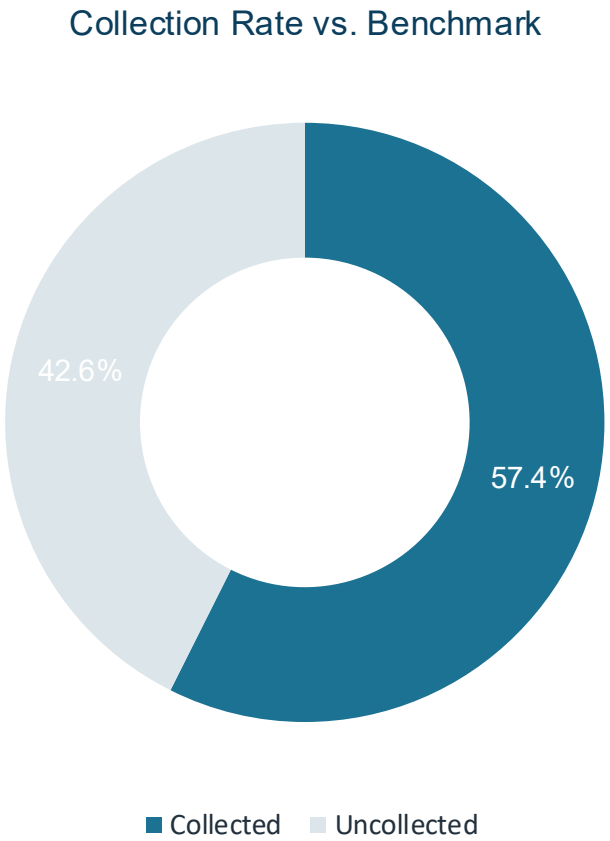
3 Findings: Generation & Collection

3,238 t/yr
Total e-waste generated across 6 districts

57.4%
Collection rate (below 70% healthy-system benchmark)

12–20 collection centers per district

Infrastructure exists at the district level, but throughput has not yet closed the collection gap.

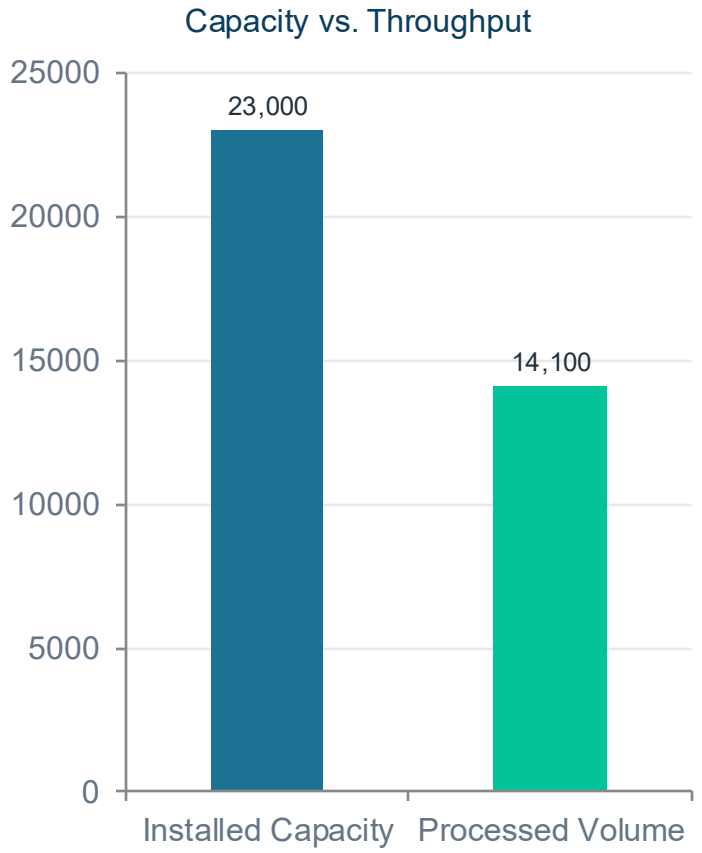


3.1 Formal Recycling & Treatment

5	23,000 t/yr	14,100 t
Authorized recyclers	Installed capacity	Processed volume

61% capacity utilization · Score: 20 / 20 — full marks against benchmark

This category is the state's clearest strength: installed capacity comfortably exceeds current throughput, giving room for collection volumes to scale without new infrastructure investment.

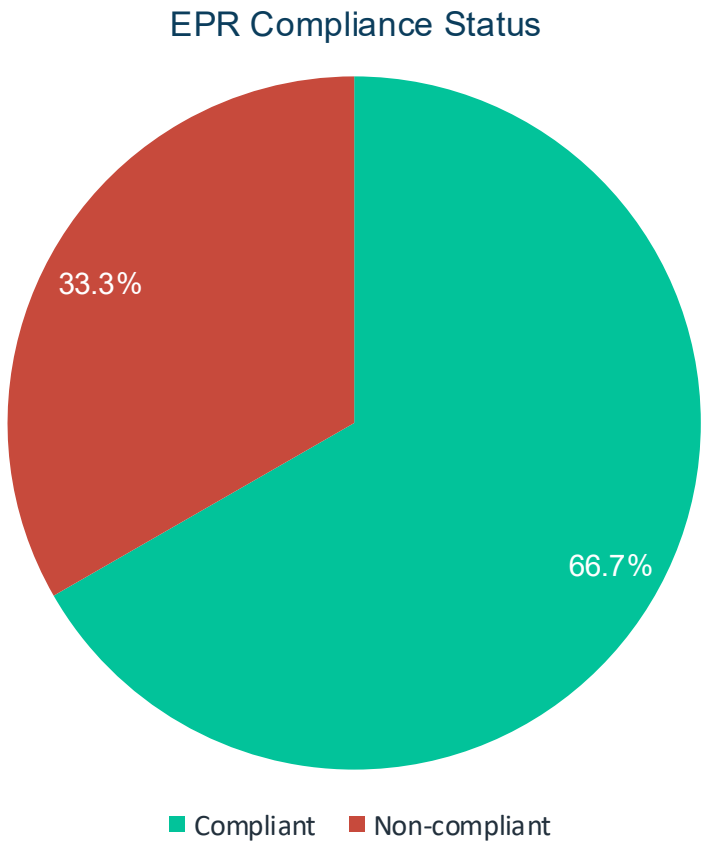


3.2 Regulatory Compliance

6
EPR entities tracked

66.7%
Fully compliant

2 penalties issued · 1 resolved to date



Score: 10 / 15

Two-thirds compliance leaves meaningful ground to cover before the state can claim full EPR conformity across all tracked entities.

3.3 Informal Sector & Environmental Safeguards

4
Known informal hotspots

25%
Weighted formalization score

3
Health camps conducted

515 residents screened · 3 of 4 sites exceed contamination limits (Pb, Cd)

[INSERT CHART: Contamination Levels by Site — Pb / Cd]

Only partial formalization has been achieved across informal recycling hotspots, and heavy-metal contamination above safe limits was confirmed at the majority of monitored sites — flagging this as a priority intervention area.

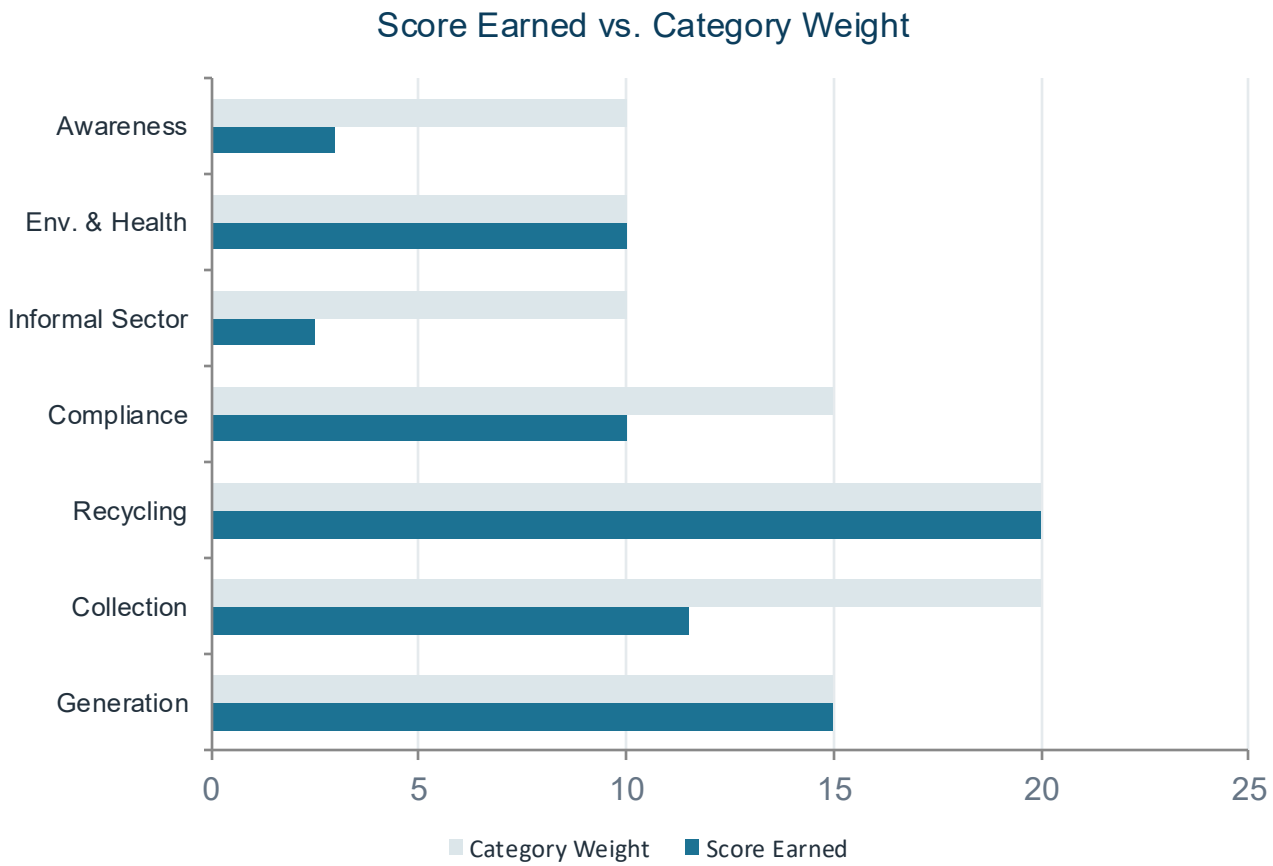
3.4 Awareness & Outreach

6	4,050	3
Programs held	People reached	Counted as formal collection drives

Score: 3 / 10 — lowest-performing category relative to its benchmark

Awareness outreach shows the largest gap between benchmark and achievement of any category, making it the single highest-leverage area for score improvement.

4 Discussion



Score Band: GOOD (65–84)

- Strongest: Formal Recycling & Treatment (20/20) and Environmental & Health Safeguards (10/10) — capacity and monitoring systems are in place and performing at benchmark.
- Weakest: Awareness & Outreach (3/10) and Informal Sector Integration (2.5/10) — the largest untapped gains for the next scoring cycle.
- Audit-then-score approach: validating formulas before publishing results prevents silent errors from masking true category performance.

5 Recommendations

1 Raise collection rates

Prioritize district-level interventions in lagging districts to close the gap to the 70% benchmark.

2 Formalize informal hotspots

Expand training and registration support to bring known hotspots into the formal system.

3 Scale awareness campaigns

Target the largest single score gap by broadening outreach beyond current program reach.

4 Maintain quarterly re-scoring

Re-run the validated scorecard each quarter to track trend and catch drift early.

6 Dashboard Preview

An interactive glassmorphism analytics dashboard built on this validated dataset — presenting the score ring, KPI cards, and category charts in a live, filterable view.

[INSERT SCREENSHOT: Live Dashboard — Score Ring, KPI Cards, Category Charts]

7

Conclusion

Current score: 72 / 100 — GOOD band, with a clear roadmap to the Excellent tier through targeted gains in awareness and informal-sector integration.

This scorecard is a reusable framework — the same seven-category methodology and weighting scheme can be applied to any Indian state, enabling consistent, comparable e-waste governance tracking nationwide.

72 / 100
Final validated score

GOOD
Performance band

Any State
Framework is fully reusable

Refs

References / Data Sources

1 E-Waste (Management) Rules, 2022. Central Pollution Control Board (CPCB), Government of India.

2 Internal validated State E-Waste Tracker workbook — primary data source (7-sheet dataset, audited).

3 Maes, T. & Preston-Whyte, F. (2022). "E-waste it wisely: lessons from Africa." SN Applied Sciences, 4:72 — cited as a comparative international case study.